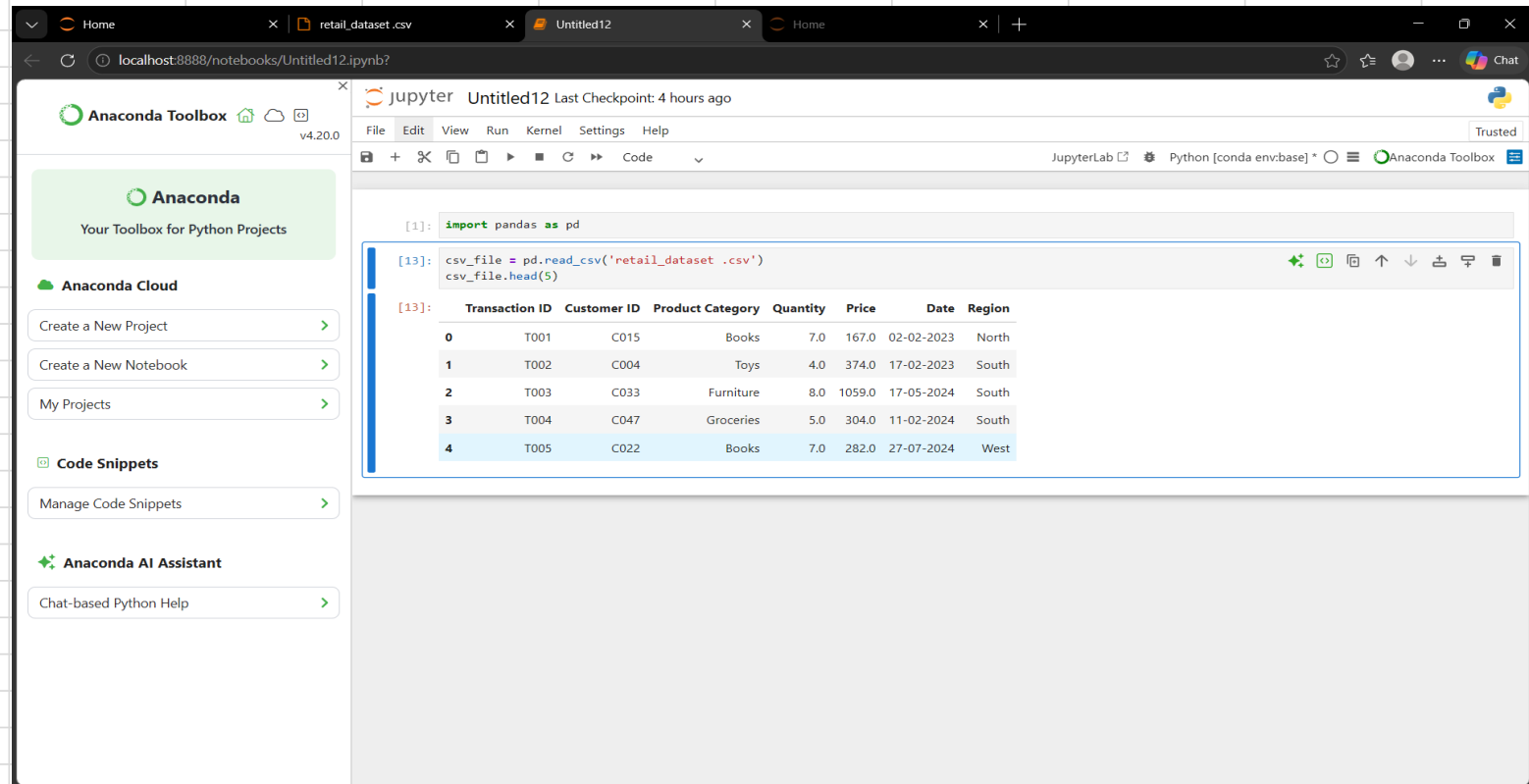


Libraries: NumPy & Pandas

QUESTION 1

Load the given retail sales dataset into a Pandas DataFrame and display the first 5 rows along with the dataset information to understand its structure.

ANSWER



The screenshot shows a JupyterLab interface with a notebook titled 'Untitled12'. The left sidebar displays the Anaconda Toolbox, Anaconda Cloud, Code Snippets, and Anaconda AI Assistant. The main area shows the notebook content, which includes the following code and output:

```
[1]: import pandas as pd
```

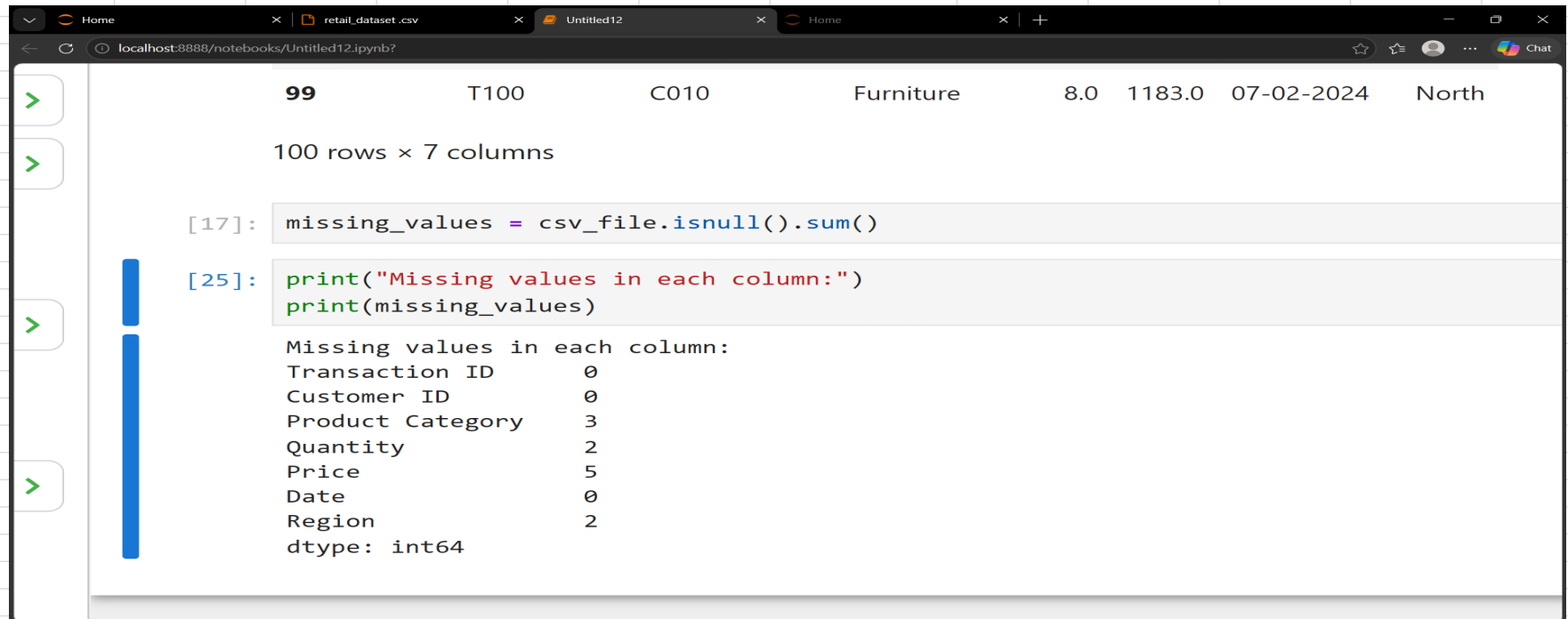
```
[13]: csv_file = pd.read_csv('retail_dataset .csv')
      csv_file.head(5)
```

The output of the code is a table showing the first 5 rows of the dataset:

	Transaction ID	Customer ID	Product Category	Quantity	Price	Date	Region
0	T001	C015	Books	7.0	167.0	02-02-2023	North
1	T002	C004	Toys	4.0	374.0	17-02-2023	South
2	T003	C033	Furniture	8.0	1059.0	17-05-2024	South
3	T004	C047	Groceries	5.0	304.0	11-02-2024	South
4	T005	C022	Books	7.0	282.0	27-07-2024	West

QUESTION 2

Inspect the dataset and identify missing values in each column. Display the total number of missing values present in the dataset.

ANSWER

The screenshot shows a Jupyter Notebook running in a web browser. The browser tabs include 'Home', 'retail_dataset.csv', 'Untitled12', and another 'Home'. The address bar shows 'localhost:8888/notebooks/Untitled12.ipynb?'. The notebook interface has a left sidebar with green arrow icons. The main area displays a preview of a dataset with 100 rows and 7 columns. The first row of data is: 99, T100, C010, Furniture, 8.0, 1183.0, 07-02-2024, North. Below the preview, the text '100 rows x 7 columns' is shown. Two code cells are visible. The first cell, labeled '[17]:', contains the code 'missing_values = csv_file.isnull().sum()'. The second cell, labeled '[25]:', contains the code 'print("Missing values in each column:"); print(missing_values)'. The output of the second cell is displayed below the code: 'Missing values in each column:', 'Transaction ID 0', 'Customer ID 0', 'Product Category 3', 'Quantity 2', 'Price 5', 'Date 0', 'Region 2', and 'dtype: int64'.

Transaction ID	Customer ID	Product Category	Quantity	Price	Date	Region	
99	T100	C010	Furniture	8.0	1183.0	07-02-2024	North

```
[17]: missing_values = csv_file.isnull().sum()

[25]: print("Missing values in each column:")
      print(missing_values)

Missing values in each column:
Transaction ID      0
Customer ID         0
Product Category    3
Quantity            2
Price               5
Date                0
Region              2
dtype: int64
```

QUESTION 3

Replace missing values in the Quantity and Price columns with their respective mean values using Pandas.

ANSWER

The screenshot shows an Anaconda Jupyter Notebook interface. The left sidebar contains the 'Anaconda' logo, 'JupyterLab for Python Projects', and a 'Cloud' section with buttons for 'Project', 'Notebook', and 'Environments'. Below this is an 'AI Assistant' section with a 'Non Help' button. The main notebook area displays a code cell with the following Python code:

```
[63]: csv_file["Quantity"].fillna(csv_file["Quantity"].median(), inplace=True)
      csv_file["Price"].fillna(csv_file["Price"].median(), inplace=True)
      print(csv_file)
```

The output of the code is a preview of the 'retail_dataset.csv' file. It shows the first 100 rows of the dataset, which has 7 columns: Transaction ID, Customer ID, Product Category, Quantity, Price, and Date. The data is displayed in a table format with alternating light blue and white rows. The first 10 rows are shown, followed by an ellipsis, and then the last 10 rows (95-100). The last row (100) shows a missing value in the Quantity column (8.0) and a missing value in the Price column (1183.0).

Transaction ID	Customer ID	Product Category	Quantity	Price	Date
0	T001	C015	Books	7.0	167.0
1	T002	C004	Toys	4.0	374.0
2	T003	C033	Furniture	8.0	1059.0
3	T004	C047	Groceries	5.0	304.0
4	T005	C022	Books	7.0	282.0
...	...	...	...	...	...
95	T096	C036	Furniture	10.0	2901.0
96	T097	C021	Clothing	9.0	1077.0
97	T098	C015	Clothing	7.0	643.0
98	T099	C005	Groceries	9.0	429.0
99	T100	C010	Furniture	8.0	1183.0

The output also shows the first 100 rows of the 'Region' column, which has 7 categories: North, South, West, and East. The first 10 rows are shown, followed by an ellipsis, and then the last 10 rows (95-100). The last row (100) shows a missing value in the Region column (North).

```
[100 rows x 7 columns]
```

QUESTION 4

Remove all rows where the Product Category or Region column contains missing values.

ANSWER

The screenshot displays a Jupyter Notebook environment within the Anaconda Toolbox. The notebook is titled 'Untitled12' and shows a code cell [65] that filters a CSV file to remove rows with missing values in the 'Product Category' or 'Region' columns. The output of the code is a DataFrame with 95 rows and 7 columns, displaying transaction data.

```
[65]: csv_file = csv_file.dropna(subset=["Product Category", "Region"])
      print(csv_file)
```

	Transaction ID	Customer ID	Product Category	Quantity	Price	Date
0	T001	C015	Books	7.0	167.0	02-02-2023
1	T002	C004	Toys	4.0	374.0	17-02-2023
2	T003	C033	Furniture	8.0	1059.0	17-05-2024
3	T004	C047	Groceries	5.0	304.0	11-02-2024
4	T005	C022	Books	7.0	282.0	27-07-2024
..	..	..	..	..	..	..
95	T096	C036	Furniture	10.0	2901.0	08-08-2023
96	T097	C021	Clothing	9.0	1077.0	19-07-2024
97	T098	C015	Clothing	7.0	643.0	13-08-2023
98	T099	C005	Groceries	9.0	429.0	20-01-2024
99	T100	C010	Furniture	8.0	1183.0	07-02-2024

The output also shows the 'Region' column for each row, with values: North, South, South, South, West, .., North, North, North, South, North.

```
Region
0    North
1    South
2    South
3    South
4    West
..    ..
95   North
96   North
97   North
98   South
99   North
```

[95 rows x 7 columns]

QUESTION 5

Create a new column called Revenue using NumPy, where:
 $\text{Revenue} = \text{Quantity} \times \text{Price}$
Display the updated dataset with the new column.

ANSWER

The screenshot shows a JupyterLab interface with a browser window at the top displaying 'localhost:8888/notebooks/Untitled12.ipynb?'. The left sidebar contains the Anaconda logo and navigation options: 'Project', 'Notebook', 'Snippets', 'AI Assistant', and 'Python Help'. The main area shows a code cell [69] with the following Python code:

```
[69]: import numpy as np
      csv_file["Revenue"] = np.multiply(csv_file["Quantity"], csv_file["Price"])
      print(csv_file)
```

Below the code, the output of the code cell is displayed, showing the updated dataset with 95 rows and 8 columns. The output is a table with the following structure:

	Transaction ID	Customer ID	Product Category	Quantity	Price	Date	\
0	T001	C015	Books	7.0	167.0	02-02-2023	
1	T002	C004	Toys	4.0	374.0	17-02-2023	
2	T003	C033	Furniture	8.0	1059.0	17-05-2024	
3	T004	C047	Groceries	5.0	304.0	11-02-2024	
4	T005	C022	Books	7.0	282.0	27-07-2024	
..	...	...	...	...	...	...	...
95	T096	C036	Furniture	10.0	2901.0	08-08-2023	
96	T097	C021	Clothing	9.0	1077.0	19-07-2024	
97	T098	C015	Clothing	7.0	643.0	13-08-2023	
98	T099	C005	Groceries	9.0	429.0	20-01-2024	
99	T100	C010	Furniture	8.0	1183.0	07-02-2024	

Below the first table, a second table is shown, which is a summary of the 'Revenue' column:

	Region	Revenue
0	North	1169.0
1	South	1496.0
2	South	8472.0
3	South	1520.0
4	West	1974.0
..	...	...
95	North	29010.0
96	North	9693.0
97	North	4501.0
98	South	3861.0
99	North	9464.0

The output is labeled as [95 rows x 8 columns] at the bottom.

QUESTION 6

Using NumPy, calculate the total revenue generated from all transactions in the dataset.

ANSWER

The screenshot shows a JupyterLab interface with a code cell containing the following Python code:

```
[69]: import numpy as np
      csv_file["Revenue"] = np.multiply(csv_file["Quantity"], csv_file["Price"])
      print(csv_file)
```

The output of the code is a table with 100 rows (indices 0 to 99) and 8 columns. The first 5 rows are shown as an example:

	Transaction ID	Customer ID	Product Category	Quantity	Price	Date	
0	T001	C015	Books	7.0	167.0	02-02-2023	
1	T002	C004	Toys	4.0	374.0	17-02-2023	
2	T003	C033	Furniture	8.0	1059.0	17-05-2024	
3	T004	C047	Groceries	5.0	304.0	11-02-2024	
4	T005	C022	Books	7.0	282.0	27-07-2024	

The table continues with rows 95 to 99, showing the final revenue calculation for each region:

	Region	Revenue
0	North	1169.0
1	South	1496.0
2	South	8472.0
3	South	1520.0
4	West	1974.0
95	North	29010.0
96	North	9693.0
97	North	4501.0
98	South	3861.0
99	North	9464.0

QUESTION 7

Group the dataset by Product Category and calculate the total revenue generated by each category.

ANSWER

The screenshot shows an Anaconda Jupyter Notebook interface. The left sidebar contains the Anaconda logo, a search bar, and navigation links for 'New Project', 'New Notebook', 'Snippets', and 'AI Assistant'. The main area displays a Jupyter Notebook with the following content:

```
..      ...      ...      ...      ...      ...      ...
95      T096      C036      Furniture      10.0      2901.0      08-08-2023
96      T097      C021      Clothing      9.0      1077.0      19-07-2024
97      T098      C015      Clothing      7.0      643.0      13-08-2023
98      T099      C005      Groceries      9.0      429.0      20-01-2024
99      T100      C010      Furniture      8.0      1183.0      07-02-2024
```

Below the data, a summary of the dataset is shown:

```
Region      Revenue
0      North      1169.0
1      South      1496.0
2      South      8472.0
3      South      1520.0
4      West      1974.0
..      ...      ...
95      North      29010.0
96      North      9693.0
97      North      4501.0
98      South      3861.0
99      North      9464.0
```

The output of the code is displayed as a table:

Product Category	Revenue
Beauty	27647.000000
Books	29597.938776
Clothing	85006.000000
Electronics	93204.315789
Furniture	129664.526316
Groceries	11978.000000
Sports	34484.000000
Toys	53947.879699

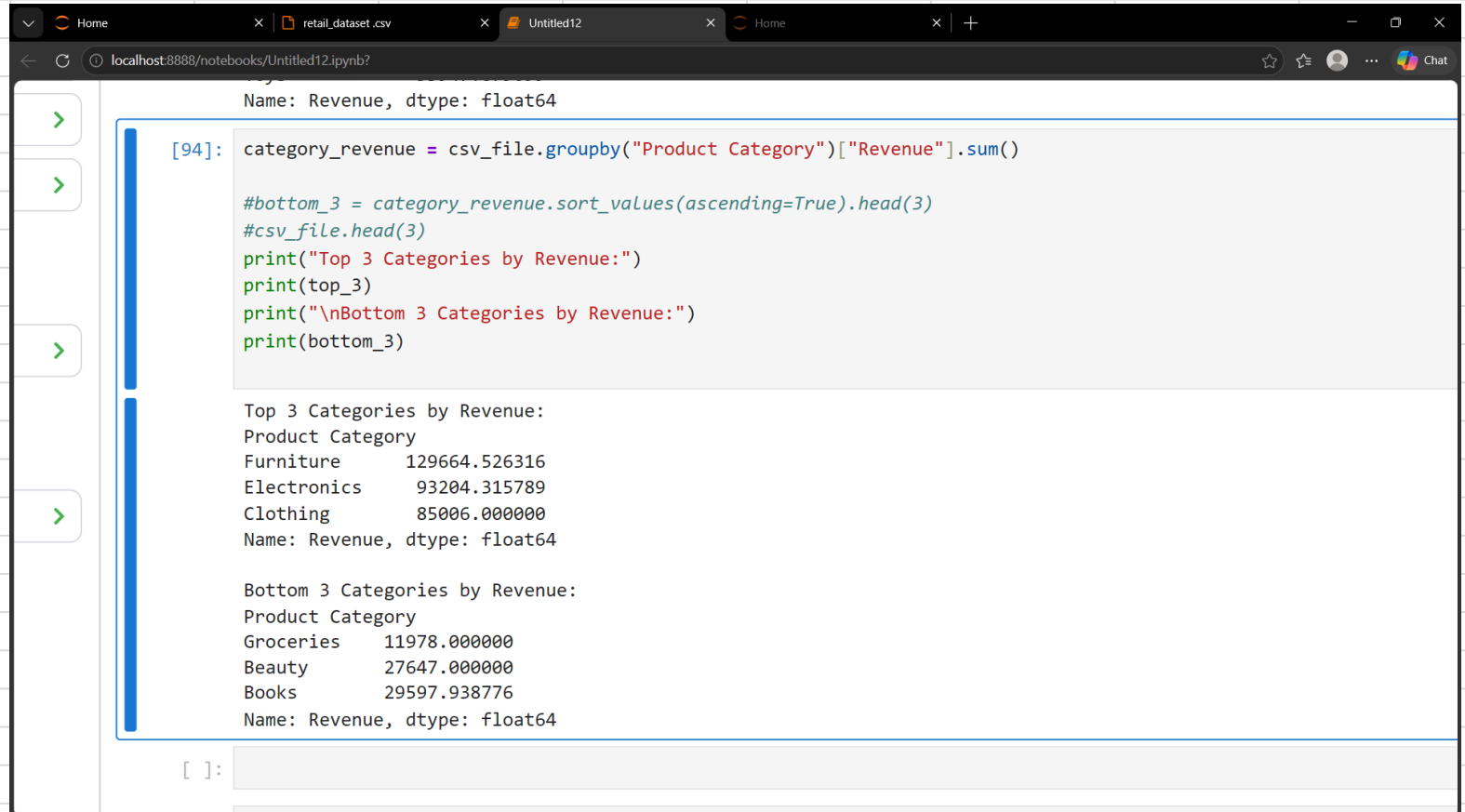
The code executed in the notebook is:

```
[74]: csv_file["Revenue"] = np.multiply(csv_file["Quantity"], csv_file["Price"])
category_revenue = csv_file.groupby("Product Category")["Revenue"].sum()
print(category_revenue)
```

QUESTION 8

Based on the revenue calculated for each product category, identify the top 3 categories with the highest revenue and the bottom 3 categories with the lowest revenue.

ANSWER



```
Name: Revenue, dtype: float64

[94]: category_revenue = csv_file.groupby("Product Category")["Revenue"].sum()

      #bottom_3 = category_revenue.sort_values(ascending=True).head(3)
      #csv_file.head(3)
      print("Top 3 Categories by Revenue:")
      print(top_3)
      print("\nBottom 3 Categories by Revenue:")
      print(bottom_3)

Top 3 Categories by Revenue:
Product Category
Furniture      129664.526316
Electronics    93204.315789
Clothing       85006.000000
Name: Revenue, dtype: float64

Bottom 3 Categories by Revenue:
Product Category
Groceries      11978.000000
Beauty         27647.000000
Books          29597.938776
Name: Revenue, dtype: float64

[ ]:
```

QUESTION 9

Group the dataset by Region and calculate the total revenue generated in each region. Identify the region with the highest revenue and the region with the lowest revenue.

ANSWER

The screenshot shows a Jupyter Notebook window with the following content:

```
retail_dataset.csv  Untitled12
```

```
localhost:8888/notebooks/Untitled12.ipynb
```

```
groceries 11978.000000
Beauty    27647.000000
Books     29597.938776
Name: Revenue, dtype: float64
```

```
[96]: region_revenue = csv_file.groupby("Region")["Revenue"].sum()
print("Revenue by Region:")
print(region_revenue)
highest_region = region_revenue.idxmax()
highest_value = region_revenue.max()
lowest_region = region_revenue.idxmin()
lowest_value = region_revenue.min()
print("\nRegion with Highest Revenue:")
print(highest_region, "=", highest_value)
print("\nRegion with Lowest Revenue:")
print(lowest_region, "=", lowest_value)
```

```
Revenue by Region:
Region
East      126232.044039
North     153244.353383
South     104855.736842
West       81197.526316
Name: Revenue, dtype: float64
```

```
Region with Highest Revenue:
North = 153244.35338345866
```

```
Region with Lowest Revenue:
West = 81197.52631578947
```

QUESTION 10

Extract the month from the Date column using Pandas and group the dataset by month to calculate the total monthly revenue. Also, calculate the mean, median, and standard deviation of the Revenue column using NumPy.

ANSWER

Home

retail_dataset.csv

Untitled12

Home

localhost:8888/notebooks/Untitled12.ipynb?

Chat

KeyError: 'Date_Time'

Fix Code

[158]:

csv_file.describe()

[158]:

	Quantity	Price	Revenue
count	95.000000	95.000000	95.000000
mean	5.925456	820.032687	4900.312217
std	2.885170	595.943664	4746.217975
min	1.000000	92.000000	285.000000
25%	3.000000	429.000000	1668.000000
50%	6.000000	732.000000	3282.105263
75%	8.000000	973.000000	7352.368421
max	10.000000	2901.000000	29010.000000